



Mixed practice

1 Work out the value of:

a $\left(\frac{1}{9}\right)^2$ b $\left(\frac{1}{9}\right)^{-\frac{1}{2}}$ c $25^{-\frac{1}{2}}$ d $\left(\frac{1}{27}\right)^{-\frac{1}{3}}$

e $\left(\frac{9}{16}\right)^{\frac{1}{2}}$ f $\left(\frac{25}{100}\right)^{-\frac{1}{2}}$ g $\left(\frac{8}{125}\right)^{-\frac{1}{3}}$

2 Simplify:

a $(49x^6)^{\frac{1}{2}}$ b $(64x^3)^{-\frac{1}{3}}$ c $(4x^4)^{-\frac{1}{2}} \times 3x$

3 Work out the value of:

a $10000^{\frac{3}{4}}$ b $1000^{\frac{2}{3}}$ c $27^{-\frac{2}{3}}$ d $9^{-\frac{3}{2}}$

4 Find the value of each of these expressions.

Leave your answers as rational numbers.

a $\left(\frac{1}{8}\right)^{-\frac{2}{3}}$ b $\left(\frac{4}{9}\right)^{-\frac{3}{2}}$ c $\left(\frac{125}{64}\right)^{\frac{2}{3}}$ d $\left(\frac{5}{9}\right)^{\frac{3}{2}} \times \sqrt{5}$

5 Simplify these expressions completely. Leave your answers as rational powers of prime numbers.

a $64^{\frac{1}{3}} \times 64^{\frac{1}{12}}$ b $81^{\frac{1}{4}} \times 9^{\frac{1}{8}}$ c $\frac{16^{\frac{5}{12}}}{16^{\frac{1}{6}}}$ d $\frac{36^{\frac{3}{4}}}{\frac{1}{6^{\frac{1}{3}}}}$

6 Simplify:

a $25^2 \times 5^{-2}$ b $16^{\frac{1}{2}} \times 4^{-\frac{3}{2}}$ c $7^{-\frac{1}{2}} \times 343^{\frac{1}{2}}$

7 Simplify these expressions completely.

Leave your answers as simplified radicals.

a $\sqrt[3]{16} \times \sqrt[3]{8}$ b $\sqrt[4]{12} \times \sqrt[4]{48}$ c $\frac{\sqrt[4]{64}}{\sqrt[4]{4}}$

d $\frac{3\sqrt[3]{9}}{\sqrt[3]{81}}$ e $\sqrt[4]{4} \times \sqrt[8]{4} \times \sqrt[4]{2}$ f $\sqrt{8} \times \sqrt[3]{8} \times \sqrt[6]{8}$

g $\frac{\sqrt[4]{12}}{\sqrt[3]{12}}$ h $\frac{\sqrt[3]{9}}{\sqrt[4]{27}}$

Problem solving

8 Write down an equivalent expression that contains fractional exponents:

a 4 b 3 c $\frac{1}{4}$ d $\frac{3}{2}$

9 Write as simplified radicals with prime power radicands:

a $2\sqrt[3]{16}$ b $\sqrt{24} \times \sqrt[4]{36}$ c $\frac{4^2}{(\sqrt{8})^3}$ d $\frac{\sqrt[3]{5}}{2\sqrt[4]{10}}$

Problem solving

10 Solve these equations:

a $8^{-x} = \frac{1}{2}$ b $32^{\frac{x}{5}} = 4$ c $\left(\frac{x}{64}\right)^{\frac{2}{3}} = \frac{9}{16}$

11 Simplify, leaving your answers as simplified radicals of prime numbers:

a $25^{0.4} \times 5$ b $125^{0.3} \times 5^{0.1}$ c $9^{0.4} \times 81^{0.2} \times 3^{0.4}$
d $\frac{5^{0.9}}{125^{0.3}}$ e $\frac{12^3 \cdot 9^{0.2}}{3^{0.4}}$

12 The quotation below has been encoded using simplified surds and fractional exponents.

Without using a calculator, **simplify** the following expressions completely.

Use your results and the key to decode a quote by Stefan Banach.

Key: $2\sqrt{3} = L$, $\frac{1}{2} = I$, $2 = A$, $5\sqrt{2} = O$,
 $3 = M$, $8 = D$, $\sqrt{3} = S$, $1 = N$

'Mathematics

$$\left(\frac{1}{32}\right)^{\frac{1}{5}}$$

$$\left(\frac{1}{27^{\frac{1}{3}}}\right)^{-\frac{1}{2}}$$

$$\frac{15^{\frac{1}{2}} \times 20^{\frac{1}{2}}}{5\sqrt{3}}$$

$$\frac{\sqrt{18}\sqrt{10}}{\sqrt{4}\sqrt{15}}$$

$$\left(\frac{10^2}{2}\right)^{\frac{1}{2}}$$

$$12^{\frac{1}{2}}$$

$$\frac{64^{\frac{2}{3}}}{\sqrt[3]{8}}$$

$$\frac{3 \times \sqrt[3]{64}}{\sqrt{2}\sqrt{18}}$$

$$\frac{34 \times 64}{\frac{1}{24}}$$

$$81^{\frac{1}{4}}$$

$$\frac{64^{\frac{1}{2}}}{\sqrt[4]{256}}$$

$$\left(\frac{\frac{1}{144^{\frac{1}{3}}} \times \frac{1}{6^{\frac{1}{2}}}}{6 \times 2^{\frac{1}{3}}}\right)^{-\frac{1}{2}}$$

Solutions:

Mixed practice

1 a $\frac{1}{3}$ **b** 3 **c** $\frac{1}{5}$ **d** 3

e $\frac{3}{4}$ **f** 2 **g** $\frac{5}{2}$

2 a $7x^3$ **b** $\frac{1}{4x}$ **c** $\frac{3}{2x}$

3 a 1000 **b** 100 **c** $\frac{1}{9}$ **d** $\frac{1}{27}$

4 a 4 **b** $\frac{27}{8}$ **c** $\frac{25}{16}$ **d** $\frac{25}{27}$

5 a $2^{\frac{5}{2}}$ **b** $3^{\frac{5}{4}}$ **c** 2 **d** $2^{\frac{7}{3}} \times 3^{\frac{7}{3}}$

6 a 25 **b** $\frac{1}{2}$ **c** 7

7 a $\sqrt[3]{2^7}$ **b** $2\sqrt{6}$ **c** 2 **d** $\sqrt[3]{3}$

e 2 **f** 8 **g** $\frac{1}{\sqrt[12]{12}}$ **h** $\frac{1}{\sqrt[12]{3}}$

8 Student's own answers

9 a $\sqrt[3]{2^7}$ **b** 12 **c** $\frac{1}{\sqrt{2}}$ **d** $\frac{\sqrt[12]{5}}{\sqrt[4]{2^5}}$

10 a $x = \frac{1}{3}$ **b** $x = 2$ **c** $x = 27$

11 a $\sqrt[5]{5^9}$ **b** 5 **c** 9 **d** 1

e $2^6 \times 3^3 = 1728$

12 Mathematics is as old as man.